

SIDDH MEHTA

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EDUCATION

B.Tech in Mechanical Engineering, Indian Institute of Technology Jodhpur

2023–2027

CGPA: 8.16/10

Relevant Coursework: Fluid Mechanics, Heat Transfer, Design of Machine Elements, Kinematics & Dynamics of Machines, Production & Operations Management, Smart Manufacturing, Computer Integrated Manufacturing, Introduction to Robotics, AI/ML & Big Data for Mechanics and Design

SKILLS

CAD & Design	SolidWorks (Advanced), Autodesk Fusion 360, Blender, AutoCAD
Simulation	ANSYS (FEA & CFD), ADAMS, AVL, Gazebo, MATLAB
Robotics & AI/ML	ROS 2, Inverse Kinematics, Reinforcement Learning, MediaPipe, PyTorch, Scikit-Learn
Embedded & Hardware	PX4, iNav, Arduino, STM32, PWM/ESC control, UART, I2C, SPI, GPS, IMU integration
Programming	Python, C++, MATLAB, PySCIPOpt (column generation), Pandas, NumPy
Fabrication	SLS & FDM 3D Printing, DFAM, CNC, CAM, G-code, Composites, Weldments
Tools	Git, Ubuntu, ROS 2, QGroundControl, PlotJuggler, MS-Office, Google Colab
Soft Skills	Team leadership, Budgeting, Procurement, Vendor management, Technical documentation, Project scheduling

EXPERIENCE

Research Intern — Operations Research & Optimisation

Jun 2026 – Aug 2026

IIT BHU

Under Dr. Lakshay

- Developed a generic bus scheduling algorithm for large-scale public transit networks (project under NDA; client includes major Indian state transport undertakings).
- Designed and implemented a **column generation-based optimisation solver** in Python using PySCIPOpt to generate complete daily timetables — determining route assignment, departure times, and dwell times for entire bus fleets.
- Built a **structured GUI-driven input pipeline** that guides operators through entering network topology, demand history, and fleet constraints, then formats data into a validated input file for the solver.
- Designed a **generic output formatter** that translates solver results into customer-specific timetable formats, enabling the tool to serve different transport clients without code changes.

Team Manager / Vice Captain — SAE SUPRA India 2026

Oct 2025 – Sep 2026

IIT Jodhpur

- Led **chassis, steering, and suspension system design** in SolidWorks — geometry layout, component sizing, and design decisions for the competition vehicle.
- Managed all non-build operations for a 40-member team: budgeting, procurement, vendor relations, logistics, competition documentation (DR, ER, CR), and project scheduling.
- Trained and managed a 50–60 member first-year trainee division in CAD, simulation, and design fundamentals.
- Coordinated subsystem integration across chassis, powertrain, aerodynamics, and electronics.

Simulation & Mechanical Design Intern

Jun 2025 – Oct 2025

IndiFlo

Bengaluru, India

- Investigated PX4–Gazebo simulation fidelity for a 2-rotor VTOL tailsitter; analysed flight stack internals, actuator mappings, and motor outputs using QGroundControl and ROS 2-PlotJuggler.

- Modelled airframe geometry in **SolidWorks** and **Blender**; generated aerodynamic coefficients via AVL for feasibility analysis.
- Developed **custom Gazebo C++ plugins**, SDF updates, and Python/Colab pipelines to build partial simulation infrastructure; documented physics and engine limitations.

Mechanical Design Intern

Alt Carbon

Jan 2025 – Mar 2025

Kolkata, India

- Designed and prototyped field-deployable mechanisms for basalt-powder distribution using **SolidWorks CAD-driven design cycles**; eliminated clogging failures through iterative refinement.
- Validated performance through remote testing; improved ergonomics, manufacturability, and reliability across successive revisions.
- Produced detailed fabrication drawings and technical documentation for manufacturing hand-off.

PROJECTS

Autonomous Mini Interceptor Drones using Reinforcement Learning *B.Tech Project — IIT Jodhpur*

Developing autonomous interception behaviour for mini drone swarms using RL. Designing policy training pipelines and simulation environments for multi-agent pursuit-evasion scenarios. Under supervision of Prof. Roshan Kumar Hota.

Topology Optimised Drone Frame

SolidWorks, SIMP, FEA, SLS 3D Printing

Applied SIMP-based topology optimisation in SolidWorks achieving **34.6% mass reduction** under real structural load constraints. Validated with FEA; manufactured via SLS 3D printing — full design-to-manufacture cycle.

6-DOF Gesture-Controlled Robot Arm

ROS 2, Python, C++, Gazebo, MediaPipe, IK

Built ROS 2 package to control a Franka Panda arm via real-time hand-gesture computer vision. Implemented dynamic **Inverse Kinematics** mapping hand XYZ and wrist rotation to joint trajectories; integrated pinch-to-grasp gripper control via MediaPipe landmark detection.

Fixed-Wing Reconnaissance UAV

iNav, STM32, Matek F411 WTE, GPS, VTX

Built and field-tested a fixed-wing UAV from scratch with live video streaming and GPS navigation. Programmed STM32-based flight controller using iNav; configured PWM outputs for ESC/motor control; integrated GPS and VTX over UART. Resolved GPS–video synchronisation failures through iterative debugging.

Quadrotor Motor Failure Detection

Gazebo, PX4, Python, ROS 2 — Inter-IIT Tech Meet

Wrote custom Gazebo C++ plugin to inject motor failure at runtime; built Python telemetry analysis pipeline to identify and classify failure signatures from RPM, attitude, and vibration data.

Advanced SolidWorks Training Camp (11 Weeks)

Designed and delivered a comprehensive engineering workshop covering part modelling, surfacing, FEA, CFD, CAM, G-code generation, weldments, and chassis analysis.

COMPETITIONS

- **SAE SUPRA India 2026** — Formula-style combustion vehicle competition *2025–2026*
- **Inter-IIT Tech Meet** — Quadrotor motor failure detection *2024–2025*

POSITIONS OF RESPONSIBILITY

- **General Secretary**, Boltheads – The Automobile Society, IIT Jodhpur *Jul 2025 – Present*
- **Team Manager / Vice Captain**, SAE SUPRA India 2026, IIT Jodhpur *Oct 2025 – Sep 2026*
- **Team Member**, Robotics Club, IIT Jodhpur *2024 – 2025*
- **Point of Contact**, Guest Relations, Prometeo'24, IIT Jodhpur *Jan – Feb 2024*